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Original article

Impact of rapid microbial identification on clinical outcomes in bloodstream infection: the RAPIDO randomized trial

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ABSTRACT

Objectives: Bloodstream infection has a high mortality rate. It is not clear whether laboratory-based rapid identification of the organisms involved would improve outcome.
Methods: The RAPIDO trial was an open parallel-group multicentre randomized controlled trial. We tested all positive blood cultures from hospitalized adults by conventional methods of microbial identification and those from patients randomized (1:1) to rapid diagnosis in addition to matrix-assisted desorption ionization–time of flight mass spectrometry (MALDI-TOF MS) performed directly on positive blood cultures. The only primary outcome was 28-day mortality. Clinical advice on patient management was provided to members of both groups by infection specialists.
Results: First positive blood culture samples from 8628 patients were randomized, 4312 into rapid diagnosis and 4136 into conventional diagnosis. After prespecified postrandomization exclusions, 2740 in the rapid diagnosis arm and 2810 in the conventional arm were included in the mortality analysis. There was no significant difference in 28-day survival (81.5% 2233/2740 rapid vs. 82.3% 2313/2810 conventional; hazard ratio 1.05, 95% confidence interval 0.93–1.19, p 0.42). Microbial identification was quicker in the rapid diagnosis group (median (interquartile range) 38.5 (26.7–50.3) hours after blood sampling vs. 50.3 (47.1–72.9) hours after blood sampling, $p < 0.01$), but times to effective antimicrobial therapy were no shorter (respectively median (interquartile range) 24 (2–78) hours vs. 13 (2–69) hours). There were no significant differences in 7-day mortality or total antibiotic consumption; times to resolution of fever, discharge from hospital or de-escalation of broad-spectrum therapy or 28-day *Clostridioides difficile* incidence.
Conclusions: Rapid identification of bloodstream pathogens by MALDI-TOF MS in this trial did not reduce patient mortality despite delivering laboratory data to clinicians sooner. **A. MacGowan, Clin Microbiol Infect 2020;26:1347**
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Introduction

Improving the speed and accuracy of infection diagnostics is central to best managing infection as well as addressing and mitigating the adverse effects of inappropriate antimicrobial use.

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Several methods for rapid identification of organisms in bloodstream infections (BSIs) have been described. The pathogens detected vary, as do coidentification of resistance markers and turnaround time; however, a systematic review of a range of diagnostic methods suggested that rapid diagnosis improved survival in BSI [1,2]. However, only a few of the reported studies were randomized trials or used microorganism identification by matrix-assisted desorption ionization–time of flight mass spectrometry (MALDI-TOF MS), a technology which has been widely adopted in diagnostic clinical laboratories.

BSI is a common and important cause of death in patients with infection, with an overall mortality at 30 days in the range of 15% to 25%, and up to 50% at 12 to 36 months [3–6]. Earlier information about the identity and susceptibility to treatment of potential pathogens could enable clinicians to make adjustments to empirical therapy sooner, thereby leading to earlier initiation of both appropriate and optimal therapy as well as drug de-escalation when needed. Appropriate therapy helps survival [7], and use of narrow-spectrum therapy may reduce antibiotic resistance.

We sought to test these hypotheses in a randomized controlled trial. We compared conventional diagnostic methods with rapid diagnosis (via microbial identification by MALDI-TOF MS) [8] to see whether rapid diagnosis would reduce 28-day all-cause mortality and a range of secondary outcomes.

Methods

Design and setting

The RAPIDO prospective open parallel-group (1:1) randomized controlled trial took place in seven NHS acute hospital trusts in England and Wales between July 2012 and August 2014 (Supplementary Material, Registrations and Ethics).

Participants

Adult patients admitted to hospital for NHS care and having a blood sample culture positive for bacteria or fungi were potentially eligible for inclusion, whether or not the organisms were clinically significant. The date and time of taking the first such culture-positive blood sample (date 0, time 0) marked the start of an episode of BSI.

Ethics

The National Research Ethics Committee South West approved the trial as a whole. The Ethics and Confidentiality Committee of the National Information Governance Board for Health and Social Care, on behalf of the Secretary of State for Health, approved specific aspects relating to consent and use of patient data. North Bristol NHS Trust acted as the study's sponsor.

Interventions

All blood samples were taken from patients according to normal clinical criteria in each centre and were incubated in blood culture machines. All samples that were flagged positive were tested by conventional methods, following each laboratory's standard procedures, with Gram stain and microscopy followed by culture to isolate individual species, biochemical tests for identity and reculture to test antimicrobial susceptibility (Supplementary Material, MALDI-TOF MS Testing).

In the rapid diagnosis arm, direct MALDI-TOF MS after extraction was used in addition to conventional methods (Supplementary Material, MALDI-TOF MS Testing).

In both study arms the microbiology results were communicated and discussed with the clinical teams by infection specialists who advised on further patient management. All the sites had antibiotic treatment guidelines to inform initial empirical therapy.

Outcomes and adverse events

The primary outcome was time to death within 28 days of taking the first culture-positive blood sample. Secondary clinical outcomes were time to death within 7 days, time to resolution of fever within 7 days (first record of temperature $\leq$37.5°C if maintained $\geq$2 days), length of hospital stay within 28 days and incidence of *C. difficile* infection within 28 days.

Antimicrobial therapy was considered to be appropriate if the organism was classified as susceptible to the antimicrobial prescribed and if therapy continued for at least 36 hours (Supplementary Material).

Sample size

On the basis of previous reports [9,10] the study was designed to have 80% power (with 95% confidence in a two-sided test) to detect a 4-point reduction in 28-day mortality (from the expected 20–16%) for patients with clinically significant infection (Supplementary Material).

Data collection and statistical methods

We included all eligible randomized patients, except those declining consent, in intention-to-treat analyses for 28-day and 7-day mortality, with a prespecified plan. We used Stata 13.1 or 14.0 (StataCorp, College Station, TX, USA), and we considered two-tailed $p < 0.05$ to indicate statistical significance. Unapproached survivors were excluded from other analyses as a result of lack of data. We excluded clinically nonsignificant and polymicrobial episodes from analysis of times to initiation and de-escalation of antimicrobial therapy, and we excluded one centre's data from analysis of time to reporting microbial identity because of concerns over its quality. An additional descriptive analysis, not prespecified, aimed to describe the relationship between time to provision of microbial identification and survival status.

We used Cox proportional hazard models stratified by centre for all time-to-event outcomes except provision of microbial identity. We analysed acquisition of *C. difficile* infection by logistic regression, adjusting for centre as a fixed effect and antimicrobial consumption by ordinal logistic regression of approximate quintiles without adjustment for centre. An additional descriptive account of the relationship between time to provision of microbial identification and survival status was not prespecified.

Predefined subgroup analyses were by clinical significance, by causative organism and by presence of neutropaenia.

We used complete case analysis for most outcomes because they had less than 5% missing data, the exceptions being antimicrobial consumption and subgroup analysis for neutropaenia.

In the Supplementary Material are further details of data collection, use of data and statistical methods.

Results

Patient and baseline data

Between July 2012 and August 2014 a total of 8628 samples were randomly assigned to microbial identification using MALDI-TOF MS plus conventional methods (rapid diagnosis arm) or conventional methods alone. After excluding samples that were

ineligible or from patients who declined consent, 5550 participants were included. In the rapid diagnosis arm, data of 2740 patients were available for analysis of mortality and 2197 for other outcomes, while in the conventional arm, data of 2810 patients were analysed for mortality and 2271 for other outcomes (Fig. 1).

Baseline characteristics were similar between the randomized groups (Table 1). Median (interquartile range (IQR)) age was 60 (55–80) years, 55.5% were men and 37.1% had hospital-acquired infection. The median (IQR) modified Charlson score was 3 (2–4).

At the time the blood sample was taken, median (IQR) systolic blood pressure was 120 (105–139) mm Hg. Temperature was 38.1°C (37.2–38.7°C); 67.6% (2779/4107) had fever (temperature >37.5°C). A total of 53.6% (2292/4272) had mental disorientation and 8.5% (361/4235) were ventilated. A total of 43% (1784/4148) were receiving intravenous fluids, 8.1% (399/4198) vasopressor drugs and 5.3% (221/4182) systemic corticosteroids.

Overall, 67.4% (3010/4468) of episodes were judged to be clinically significant, including all (or the vast majority) of *Escherichia coli,* methicillin-susceptible *Staphylococcus aureus* (MSSA), *Klebsiella pneumoniae, Streptococcus pneumoniae, Pseudomonas aeruginosa, Proteus mirabilis, Enterobacter* species and methicillin-resistant *S. aureus* (MRSA) isolates, while 90.3% (1108/1227) of coagulase-negative staphylococci and 88.7% (63/71) of *Corynebacterium* spp. were not considered clinically significant.

Outcomes

The primary outcome, 28-day mortality, is shown in Table 2. A Kaplan-Meier plot of survival is shown in Fig. 2. In the rapid diagnosis group, 18.5% (507/2740) of patients died by 28 days and 17.7% (497/2810) in the conventional group; the difference was not statistically significant (HR 1.05, 95% confidence interval (CI) 0.93–1.19, p 0.42). The risk of dying within 28 days was 0.9 percentage points higher in the rapid diagnosis arm (95% CI 1.2 points lower to 2.9 points higher).

Secondary outcomes are listed in Table 2. The 7-day mortality was similar in the two groups, at 9.9% (271/2740) in the rapid diagnosis arm versus 9.3% (261/2810) in the conventional arm. Total 7-day antimicrobial consumption was also similar, with median (IQR) 8 (4.3–14.3) defined daily dose in the rapid diagnosis group versus 7.8 (4.4–14.0) in the conventional group. Rates of *C. difficile* infections were low but similar in the two groups; median times to resolution of fever and discharge from hospital were identical in the rapid diagnosis and conventional groups.

The time to provision of a microbiologic identification was significantly shorter in the rapid diagnosis group (median (IQR) 38.5 (26.7–50.3) hours) compared to the conventional group (55.2 (47.0–71.0) hours) ($p < 0.0001$). In contrast, the time to provision of Gram film and antimicrobial susceptibility results was similar in the two groups (Table 3).

There was no significant difference in the proportion of patients in the conventional group (84.2%, 1243/1477) receiving broad-spectrum antimicrobial therapy at day 7 compared to patients in the rapid diagnosis group (82.7%, 1117/1351; HR 1.10, 95% CI 0.90–1.36, p 0.35) (Table 2, Supplementary Fig. S1).

Fig. 3 shows a Kaplan-Meier curve for time to initiation of appropriate antimicrobial therapy. The apparent difference between the groups was of only borderline statistical significance (HR 0.92, 95% CI 0.84–1.00, p 0.056), with the median time to appropriate therapy, surprisingly, longer for rapid diagnosis patients than

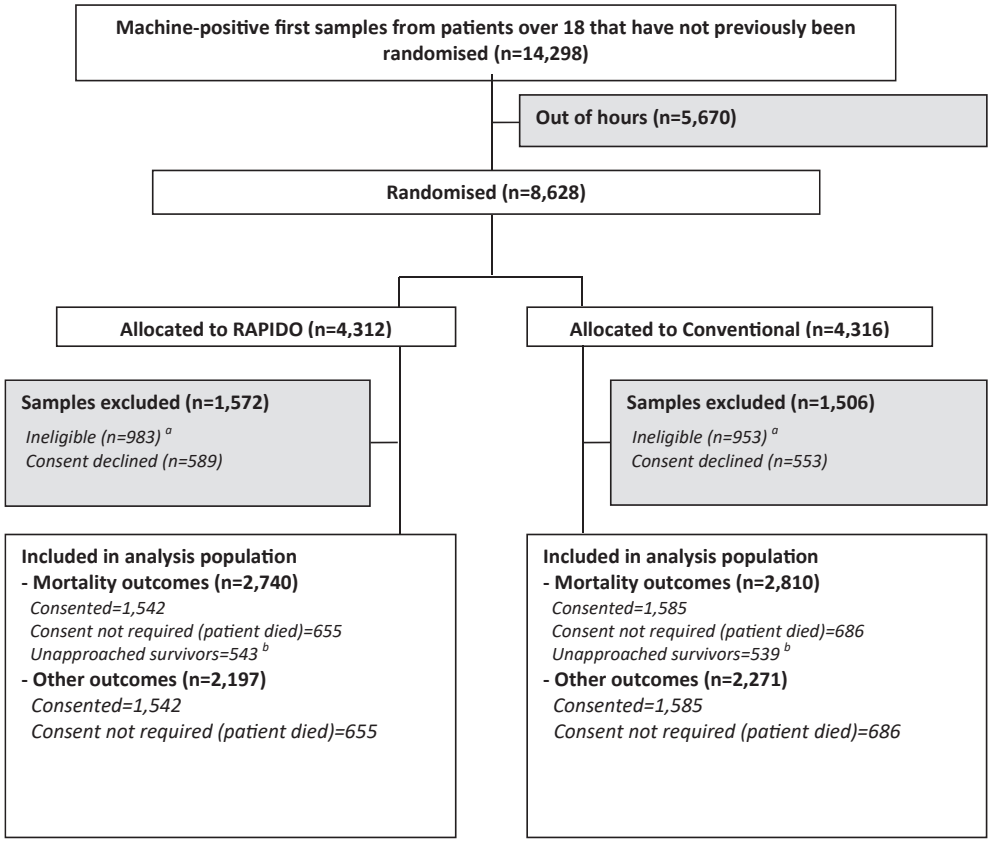


Fig. 1. Flow of patients. [a]Ineligible samples include 111 rapid diagnosis and 125 conventional samples that were randomized in error, and 872 rapid diagnosis and 828 conventional that were randomized correctly but met postrandomization exclusion criteria. [b]Unapproached survivors are eligible patients who could not be approached for consent, usually because of lack of capacity and inability to identify a consultee. They are included in mortality analysis (only) as 28-day survivors.

Table 1
Patient demographics by randomization

Characteristic	Randomized to rapid diagnosis	Randomized to conventional diagnosis
Mortality outcome analysis population ($N = 5550$)	2740	2810
Male sex	1461/2740 (53.3%)	1551/2810 (55.2%)
Age (years), median (IQR)	69 (55–80)	69 (55–82)
Other outcomes analysis population ($N = 4468$)	2197	2271
Male sex	1200/2197 (54.6%)	1281/2271 (56.4%)
Age (years), median (IQR)	69 (55–80)	69 (55–80)
Recent medical history		
Cardiac arrest in last 7 days	31/2102 (1.5%)	36/2172 (1.7%)
Chemotherapy in last month	223/2103 (10.6%)	250/2172 (11.5%)
Surgery requiring overnight stay in last 7 days	177/2104 (8.4%)	179/2171 (8.2%)
Clinical data (day −7 to 0)		
Neutrophil count (10^9/L), median (IQR)	9.2 (5.5–13.6)	9.2 (5.3–13.8)
Ventilation on day 0	177 (2078) 8.5%	184/2157 (8.5%)
Temperature (°C), median (IQR)	38.1 (37.3–38.7)	38.1 (37.2–38.7)
Fever present at time 0 (temperature >37.5°C)	1363/2013 (67.7%)	1416/2094 (67.6%)
Systolic blood pressure day 0 (mm Hg), median (IQR)[a]	120.5 (105–140)	120.0 (104–138)
Mental disorientation	1145/2102 (54.5%)	1147/2171 (52.8%)
Receiving intravenous fluids on day 10	838/2043 (41.0%)	946/2105 (44.9%)
Receiving vasopressor drugs on day 0	177/2058 (8.6%)	162/2140 (7.6%)
Systemic corticosteroids on days 0, 1 or 2	99/2049 (4.8%)	122/2133 (5.7%)
Hospital-acquired infection (>2 days after admission)	797/2101 (37.9%)	794/2182 (36.4%)
Comorbidities at day 0		
Modified Charlson Score, median (IQR)[b]	3 (2–4)	3 (2–4)
Cystic fibrosis	9/2101 (0.4%)	12/2172 (0.6%)
Prior transplant	115/2101 (5.5%)	109/2170 (5.0%)
Immunosuppressive drugs	242/2064 (11.7%)	236/2141 (11.0%)

IQR, interquartile range.
Numbers of patients with missing data (rapid diagnosis, conventional): neutrophil count 338 (156, 182); temperature 361 (184, 177); systolic blood pressure 535 (273, 262); Charlson score 1063 (510, 553).
[a] Time 0 indicates time of taking blood sample (first sample in episode).
[b] Charlson score was 0 for 484 patients (243 rapid diagnosis, 241 conventional).

for conventional (24 vs. 13 hours). There were differences between organisms, with shorter median times to appropriate therapy in the rapid diagnosis group versus the conventional group for MRSA and *P. aeruginosa* but longer times to appropriate therapy in the rapid diagnosis group for *Enterobacteriales*, coagulase-negative staphylococci, enterococci and *Candida* spp. (Supplementary Table S7).

The proportion of patients already receiving appropriate therapy before microbiologic identity became available was higher for those with MSSA, β-haemolytic streptococci, α-haemolytic streptococci or *S. pneumoniae* infections and was lower for those with enterococci, *Candida* or MRSA. Also, a higher proportion of patients were receiving appropriate therapy before the identification of

Table 2
Outcomes

Characteristic	Randomized to rapid diagnosis	Randomized to conventional diagnosis	Estimate, HR or (95% CI)	p
Mortality outcomes (including unapproached survivors) ($N = 5550$)	2740	2810		
Primary outcome: proportion died at 28 days (allowing for censoring)	18.5% (507/2740)	17.7% (497/2810)	HR 1.05 (0.93–1.19)	0.42
Secondary outcome: proportion died at 7 days (allowing for censoring)	9.9% (271/2740)	9.3% (261/2910)	HR 1.05 (0.89–1.25)	0.54
Other outcomes (excluding unapproached survivors) ($N = 4468$)	2197	2271		
Receipt of antimicrobial therapy (total DDD over 7 days): median (IQR)[a]	8.0 (4.3–14.3)	7.8 (4.4–14.0)	OR[a] 0.98 (0.89–1.09)	0.73
Clostridium difficile incidence by 28 days	27/2114 (1.3%)	30/2186 (1.4%)	OR 0.94 (0.55–1.58)	0.81
Days to resolution of fever (up to 7 days): median (IQR)	3 (1–5)	3 (1–5)	HR 0.96 (0.88–1.04)	0.32
Days to discharge from hospital (up to 28 days): median (IQR)	15 (7–NE)	15 (7–NE)	HR 0.98 (0.90–1.06)	0.55
Hours to provide information to clinical team, median (IQR) ($N = 3684$)[a]	1811	1873	—	—
For Gram film result	33.0 (23.6–44.2)	30.9 (23.3–42.8)	—	—
MALDI ID	38.5 (26.7–50.3)	—	—	<0.0001[b]
Preliminary conventional ID	—	55.2 (47.0–71.0)	—	—
Final conventional ID	—	63.8 (49.4–84.7)	—	—
Antimicrobial sensitivity results	57.2 (47.1–75.1)	57.6 (47.4–75.7)		
Outcomes for clinically significant single-organism episodes only ($N = 2828$)	1351	1477		
Proportion receiving broad-spectrum therapy at 7 days (allowing for censoring)[c]	82.7%	84.2%	HR 1.11 (0.90–1.36)	0.35
Hours to start appropriate antimicrobial therapy (up to 7 days): median (IQR)	24.0 (2.0–78.0)	13.0 (2.0–69.0)	HR 0.92 (0.84–1.00)	0.056

Numbers of patients with missing data (rapid diagnosis, conventional): receipt of antimicrobial therapy 332 (168, 164) but OR estimated with conditional mean imputation; time to discharge from hospital 146 (70, 76); Gram film result 110 (63, 47); MALDI ID (745, −); preliminary conventional ID (−, 478); final conventional ID (−, 691); earlier of conventional IDs (−, 414); antimicrobial sensitivity 1272 (688, 584).
HR, hazard ratio; IQR, interquartile range; NE, 75th percentile not estimable; OR, odds ratio.
[a] Data from one centre were excluded from analysis of information provision as a result of data quality concerns (784 patients).
[b] Wilcoxon test comparing times to MALDI ID and earlier of final and preliminary conventional ID.
[c] That is, not yet de-escalated from broad-spectrum therapy.

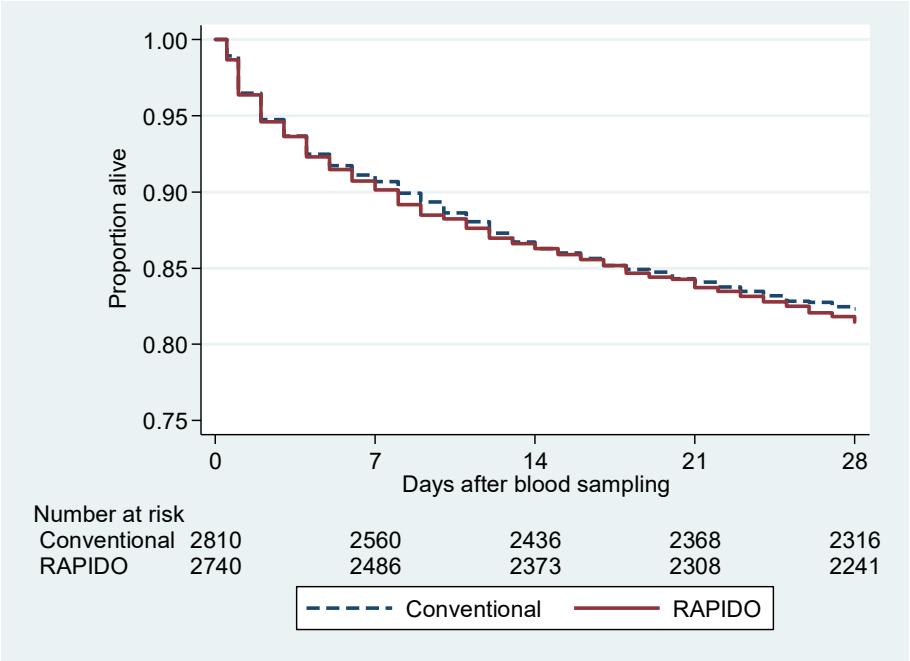


Fig. 2. Kaplan-Meier curve for 28-day mortality.

their pathogens in the conventional group than in the rapid diagnosis group (Supplementary Table S8).

There was no evidence that the effect of rapid diagnosis differed according to clinical significance for any of the outcomes (Table 3). For the primary outcome, death at 28 days, the HR (95% CI) was 1.04 (0.90–1.21) for clinically significant episodes compared to 1.11 (0.89–1.38) for non–clinically significant episodes. The subgroup analysis for 28-day mortality by the organism causing BSI showed no evidence of differing effects of rapid diagnosis for extended-spectrum β-lactamase–producing *Enterobacteriales*, MRSA, *Candida* and MSSA. There was weak evidence (p 0.071) of a differing treatment effect for infections with and without *P. aeruginosa*; on the basis of a small number of deaths (17/61 rapid diagnosis, 26/69 conventional), there was a suggestion of greater benefit (reduced mortality) of the rapid diagnosis intervention for patients with *P. aeruginosa* (HR 0.61, 95% CI 0.33–1.13) than for those without this organism (HR 1.09, 95% CI 0.96–1.23) (Fig. 4).

Discussion

To our knowledge, this rapid diagnosis randomized controlled multicentre trial is the largest yet performed of any laboratory-based rapid diagnostic method in BSI. This multicentre randomized study of rapid microorganism identification by MALDI-TOF MS after initial liquid culture showed no patient survival or other benefit compared to conventional methods.

This study provides significantly more useful information on the patient impact of MALDI-TOF MS identification as a rapid diagnostic method in BSI than the studies we included in a systematic review in 2015 [11].

More generally, the role of rapid diagnostics in BSI has been subject to a 2017 systematic review [2]. This reported data on 31 studies, most designated as pre- and postintervention designs; two were randomized controlled trials. Only four studies of MALDI-TOF MS were included, none of which was a randomized controlled trial, but all included either pharmacist- or physician-led clinical interventions that were based on the laboratory results [12–14]. Two further studies, neither of which was a randomized controlled trial, on the effect of MALDI-TOF MS identification have also been published [15,16].

The characteristics of patients in our study were similar to those described in the published literature, which characterized patients with BSI in terms of age, comorbidities and hospital-acquired infection [6,17–19]. The organisms isolated and their clinical significance were also typical [19].

Mortality at 28 days in the rapid diagnosis group (18.5%, 507/2740) was not significantly different from the conventional group (17.7%, 497/2810) (hazard ratio (HR) 1.05, 95% CI 0.93–1.19, p 0.42). Just over half of the deaths occurred in the first 7 days after blood cultures were taken (9.9%, 271/2740 rapid diagnosis; 9.3%, 261/2810 conventional). There were no differences in the secondary endpoints of total antimicrobial consumption, rates of *C. difficile* infection or time to resolution of fever; however, time to provision of a microbiologic identification of the organisms in blood cultures was reduced from a median of 56.5 hours in the conventional arm to 38.5 hours in the rapid diagnosis arm. Times to provision of Gram film results and susceptibility testing results were the same in both groups.

Four clinical trials have studied the impact of MALDI-TOF MS analysis on the outcomes of patients with BSI [12–16,20]. Three of the four were single-centre studies, and the fourth included only two centres. All four used a before-and-after study design. Huang et al. [12] used MALDI-TOF MS identification on all blood culture isolates in adults, with similar characteristics to those reported here, while Lockwood et al. [13] and Perez et al. [20] included only Gram-negative isolates and Nagel et al. [14] included only staphylococci.

In our study we were surprised to observe that the median time to start of appropriate therapy was longer overall in the rapid diagnosis group (24 hours) than the conventional group (13 hours); however, the HR of 0.92 (CI 0.84–1.0, p 0.056) was only of borderline significance. Unlike other studies, we did not observe an impact of rapid diagnosis on antimicrobial use or de-escalation of therapy [12,13]. Of importance, and not reported in other studies, is

Table 3
Outcomes for subgroups by clinical significance

Characteristic	Clinically significant episodes		Clinically nonsignificant episodes	
	Rapid diagnosis (n = 1443)	Conventional diagnosis (n = 1567)	Rapid diagnosis (n = 754)	Conventional diagnosis (n = 704)
Primary outcome: proportion died at 28 days (allowing for censoring)	23.3%	22.4%	22.8%	20.7%
Secondary outcomes				
Proportion died at 7 days (allowing for censoring)	12.8%	12.0%	11.3%	10.5%
Receipt of antimicrobial therapy (total defined daily dose over 7 days), median (IQR)	5.5 (8.8–15.9)	5.2 (8.5–15.5)	2.3 (5.7–11.7)	2.7 (6.1–11.3)
Clostridium difficile incidence by 28 days	21/1395 (1.5%)	23/1515 (1.5%)	6/719 (0.8%)	7/671 (1.0%)
Hours to provision of microbial identity (median, IQR, hours)[a]	34.0 (24.2–45.7)	53.4 (46.5–70.0)	45.7 (37.8–68.5)	66.0 (54.0–86.7)
Days to resolution of fever (up to 7 days), median (IQR)	3 (1–5)	3 (1–5)	3 (1–6)	3 (1–5)
Days to discharge from hospital (up to 28 days), median (IQR)	15 (8–NE)	15 (8–NE)	16 (6–NE)	14 (7–NE)

Total sample size of 4468 excludes unapproached survivors. Numbers of patients with missing data (clinically significant rapid diagnosis, clinically significant conventional, clinically nonsignificant rapid diagnosis, clinically nonsignificant conventional): receipt of antimicrobial therapy 332 (99, 106, 69, 58); time to microbial identity 1159 (464, 106, 281, 308); discharge from hospital 146 (44, 45, 26, 31).
IQR, interquartile range; NE, 75th percentile not estimable.
[a] One centre was excluded from analysis of time to microbial identity as a result of data quality concerns (784 patients).

the safety of rapid diagnosis with MALDI-TOF MS. We noted only 0.5% unexpected mismatches in identities between conventional and MALDI-TOF MS methods, none of which was clinically serious.

P. aeruginosa BSI was the only subgroup where rapid diagnosis appeared to have had some impact on survival (HR 0.61, 95% CI 0.33–1.13, p 0.071). However, the number of deaths in each arm was low (17/61, 28% rapid diagnosis; 26/69, 38% conventional). Fewer patients with *P. aeruginosa* received appropriate therapy compared to other pathogen groups; the median time to appropriate therapy was 26 hours with rapid diagnosis compared to 33 hours for conventional methods. This implies that MALDI-TOF MS identification may have led to earlier appropriate therapy with this pathogen group.

Comparison of our data with the published trials that involved the introduction of an Antimicrobial Stewardship Teams (AST) at the same time as the introduction of a rapid diagnostic intervention is problematic because all the hospitals in our study already had infection specialists closely linked to laboratory services. These specialists would discuss all significant blood cultures with clinical teams and guide them on both antimicrobial therapy and further investigations. Thus, we measured the single effect of adding rapid diagnosis in a setting with established AST. Hence, in most other studies in which rapid diagnosis is linked to a clinical intervention team, it is perhaps the work of the clinical intervention team which results in improved outcomes, not the rapid diagnostic tool.

What is clear from our study is that the time for information to become available from microbiologic analysis of blood cultures (median times to communicating Gram film results to clinical staff on the ward was 33 hours for rapid diagnosis and 30.9 hours for conventional diagnosis) is longer than the median times to starting appropriate antimicrobial therapy (24 and 13 hours respectively). In addition, for the large pathogen groups, median times to appropriate therapy were short in both the rapid diagnosis and conventional arms—for example, 16 and 9 hours for *Enterobacteriales*. A substantial proportion of patients with these pathogens were already receiving appropriate therapy before the Gram

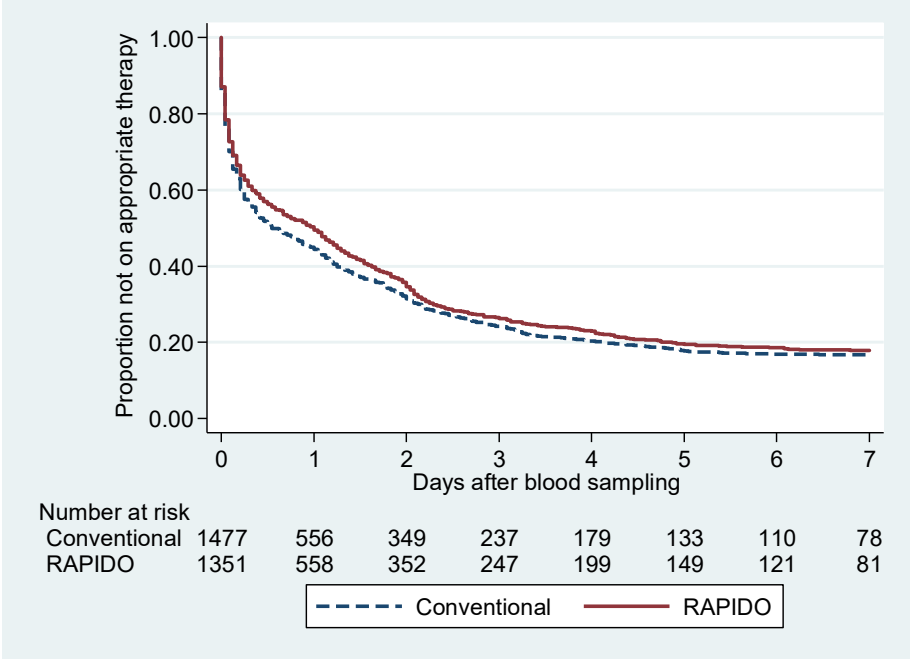


Fig. 3. Kaplan-Meier curve for time to initiation of appropriate antimicrobial therapy.

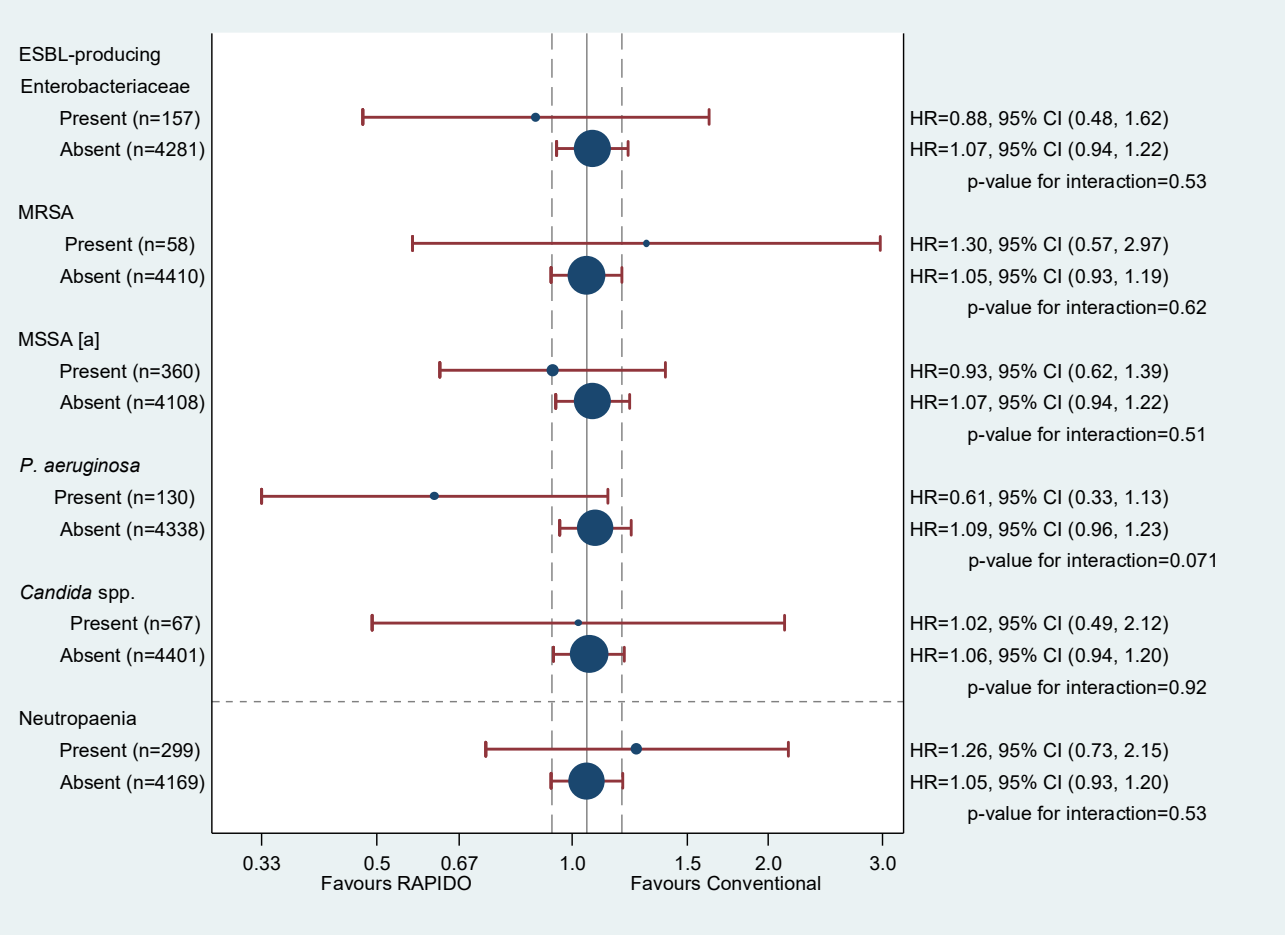


Fig. 4. Subgroup analysis for 28-day mortality by causative pathogen and neutropaenia. [a]MSSA subgroup analysis was not prespecified. CI, confidence interval; ESBL, extended-spectrum β-lactamase; HR, hazard ratio; MRSA, methicillin-resistant *Staphylococcus aureus*; MSSA, methicillin-susceptible *S. aureus*.

stain result was available. As many patients are already receiving appropriate antibiotics and there is a relatively larger risk of death in the first few days after blood cultures are taken, it is hard to show a survival benefit for rapid diagnostics that depends on prior culture of a pathogen, as this will take 18 to 24 hours. In addition, the time to availability of susceptibility results was not changed by use of MALDI-TOF MS for bacterial identification.

Our study has several limitations, notably the length of time that has elapsed since the study was conducted. MALDI-TOF MS technology has been widely introduced into laboratories for reasons other than improving the timeliness of results and improving clinical outcomes. In addition, although infection consultation teams interpreted and advised results in all the centres, there may have been differences in these as well as in the ways the laboratories worked.

In conclusion, this study of rapid microbial identification in BSI by MALDI-TOF MS directly on blood cultures after the initial culture period did not show patient survival benefit or advantages in terms of secondary endpoints. While MALDI-TOF MS produced more rapid pathogen identification, many patients in the study were already receiving appropriate antibiotics before the laboratory results became available, and many patients with BSI die soon after blood cultures are taken. This suggests that further studies on rapid diagnostics should focus on methods that do not require an initial period of culture, patient groups in which initial antimicrobial treatment is likely to be inappropriate or optimization of diagnostic pathways to further speed the provision of useful data from blood cultures.

Transparency Declaration

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cmi.2020.01.030.

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